

Automatic Speech Recognition Using Limited Vocabulary: A Survey

Algorithms Used	Advantages	Disadvantages	Limitations	Results	Conclusion
HMM	Considered a generator of acoustic vectors; Finite-state automaton with transition probabilities; Accounts for variability in speech.			Used for acoustic modeling in limited vocabulary ASR; Achieves good speech recognition results.	HMM and DNN-based AMs achieve good speech recognition results; DNN even outperform GMMs on a variety of speech recognition benchmarks.
DNN	Work well for ASR when compared with Gaussian Mixture Model-based HMMs; Outperform GMM-HMM systems in certain tasks; Derive higher-level features from lower-level ones; Less sensitive to conditions such as noise.			Used for acoustic modeling in limited vocabulary ASR; Achieves good speech recognition results; DNN even outperform GMMs on a variety of speech recognition benchmarks.	HMM and DNN-based AMs achieve good speech recognition results; DNN even outperform GMMs on a variety of speech recognition benchmarks.
Viterbi	Good for short commands, meaning for small or limited vocabulary; Seeks the most probable candidate among states in an HMM.			Used for decoding in limited vocabulary ASR.	

